

# METROPOLITAN ACADEMIC HOSPITAL - TROPICAL MEDICINE DIVISION

SPECIALIZED VECTOR-BORNE CLINICAL DIAGNOSTICS & HEMATOLOGY DIVISION

Accredited Medical Reference Laboratory • ISO 15189 Certified

|                                |                                                           |                                  |
|--------------------------------|-----------------------------------------------------------|----------------------------------|
| Patient ID: PAT-DNG-0009       | Patient Name: Ritu Islam                                  | Age / Sex: 32.0 Y / Female       |
| Contact Phone: +91 79496 63204 | Email: rituislam.009@medcare-hospital.org                 | National Aadhaar: 7742 8431 3387 |
| PAN Number: XLVXD6516Z         | Referring Consultant: Dr. Tariqul Islam, PhD, Pathologist | Sample Type: Whole Blood (EDTA)  |

## DENGUE SEROLOGY & RAPID VIROLOGICAL ASSAYS

| Test Parameter                   | Observed Result       | Biological Reference Interval | Clinical Interpretation                      |
|----------------------------------|-----------------------|-------------------------------|----------------------------------------------|
| Dengue NS1 Antigen (ELISA/Rapid) | POSITIVE              | Negative (< 0.90 Index)       | Active Dengue Viral Infection                |
| Dengue IgM Antibody              | POSITIVE (3.42 Index) | Negative (< 1.00 Index)       | Primary Immune Response Active               |
| Dengue IgG Antibody              | POSITIVE (2.81 Index) | Negative (< 1.00 Index)       | Prior Flavivirus Exposure / Secondary Dengue |

## HEMATOLOGICAL PARAMETERS & COMPLETE BLOOD COUNT (CBC)

| Hematology Parameter             | Patient Value               | Reference Range                  | Alert / Clinical Status |
|----------------------------------|-----------------------------|----------------------------------|-------------------------|
| Platelet Count (PLT)             | 143.0 x10 <sup>3</sup> / uL | 150 - 450 x10 <sup>3</sup> / uL  | Adequate                |
| White Blood Cell (WBC)           | 3.63 x10 <sup>3</sup> / uL  | 4.0 - 11.0 x10 <sup>3</sup> / uL | Leukopenia              |
| Hematocrit (HCT / PCV)           | 35.5 %                      | 36.0 - 48.0 %                    | Normal                  |
| Red Blood Cell (RBC)             | 4.1 x10 <sup>6</sup> / uL   | 3.8 - 5.5 x10 <sup>6</sup> / uL  | Standard                |
| Lymphocytes (%)                  | 15.7 %                      | 20.0 - 40.0 %                    | Normal                  |
| Neutrophils (%)                  | 67.7 %                      | 40.0 - 75.0 %                    | Normal                  |
| Alanine Aminotransferase (ALT)   | Not Done                    | 7 - 56 U/L                       | Normal                  |
| Aspartate Aminotransferase (AST) | Not Done                    | 10 - 40 U/L                      | Normal                  |

## TRIAGE CATEGORIZATION & CLINICAL IMPRESSION

|                                |                                                                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WHO Clinical Triage:           | Category A (Mild / Outpatient)                                                                                                                                                                                                            |
| Critical Findings:             | Acute Dengue Viremia (NS1 Antigen detected); Leukopenia (WBC: 3.63 k/uL - immunodepression phase)                                                                                                                                         |
| Clinical Management Directive: | Maintain strict fluid balance chart. Monitor CBC every 12-24 hours. Contraindicated: Aspirin, Ibuprofen, NSAIDs due to platelet inhibition. Recommend oral rehydration salts (ORS) or isotonic crystalloid infusion if vomiting persists. |

Electronically authenticated laboratory report.

Authorized Signatory:  
Dr. Tariqul Islam, PhD, Pathologist

**DATA NOTICE & DISCLAIMER:** This medical report contains synthetic patient identifiers generated from the Bangladesh Dengue Research Dataset. Intended exclusively for AI pair programming, testing retrieval-augmented generation pipelines, and PII masking demonstrations.